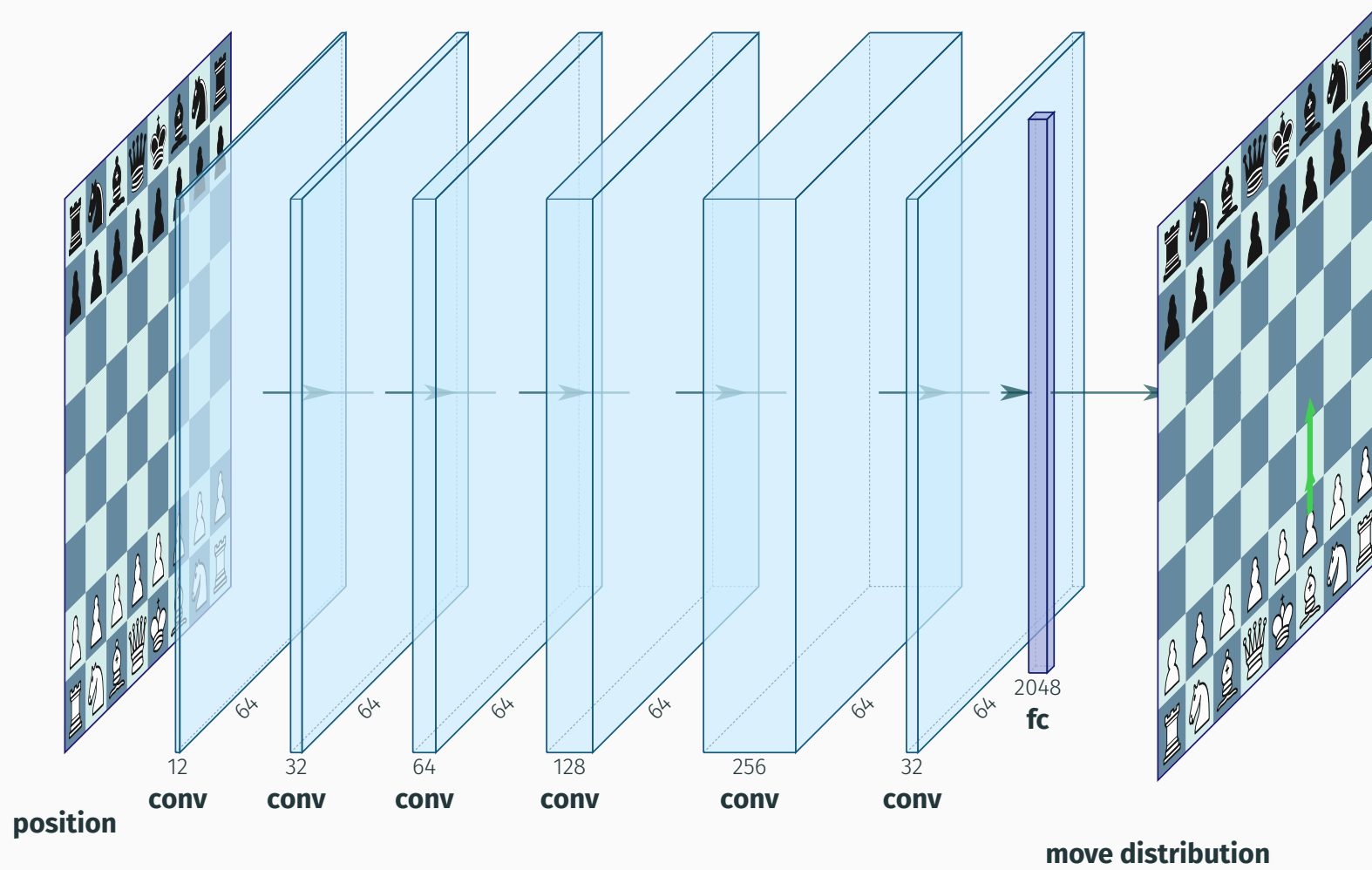


Exploring Interaction Between Behavior Cloning and Reward Shaping in Chess

Jakob Lambert-Hartmann

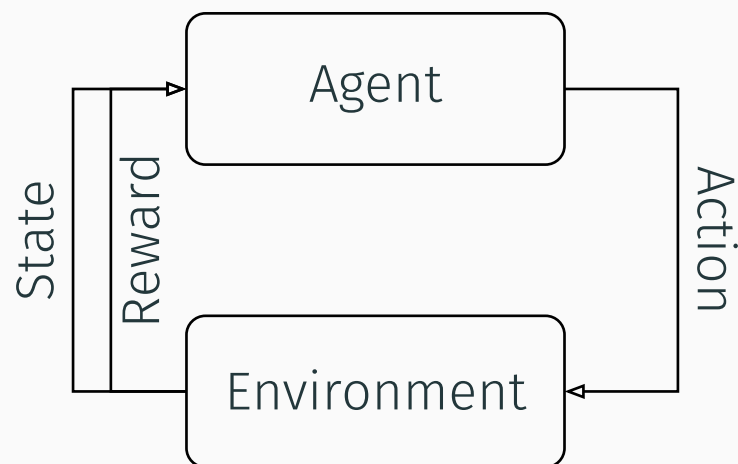
2026-02-04

Introduction



Reinforcement Learning

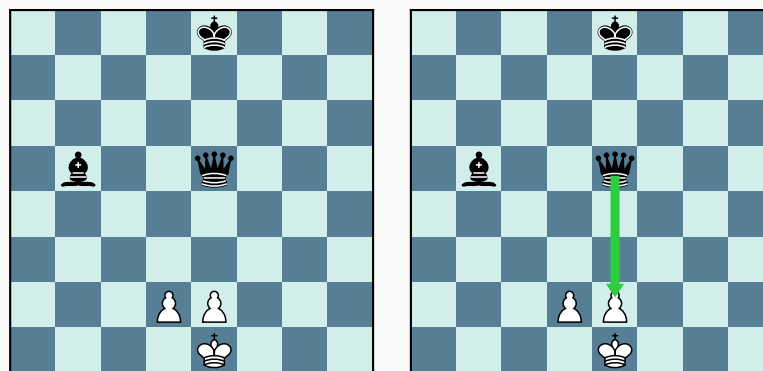
Reinforcement Learning



Problems

- Sparse rewards → poor performance
- Reward Shaping → difficult [1]:
 - Missaligned rewards
 - Reward hacking
 - Prioritizing wrong rewards

Imitation Learning: Behavior Cloning



- Supervised learning
- Agent learns expert demonstrations
- e.g. puzzles, grandmaster games

Problems

- Demonstration sample bias → poor performance [2]

Combine both approaches

1. Pretrain: Behavior Cloning
2. Finetune: Reward Based Reinforcement Learning

Can both approaches stabilize each other?

- Behavior Cloning: Stabilize RL with sparse rewards
- Reinforcement Learning: Compensates for data bias

Lets find out!

- Train different models:
 - Architecture: Fully connected, cnn, residual block
 - Pretraining Data: Heavily biased → less biased
 - Rewards Sets: Sparse → Dense

Architecture



Architecture

FCNN

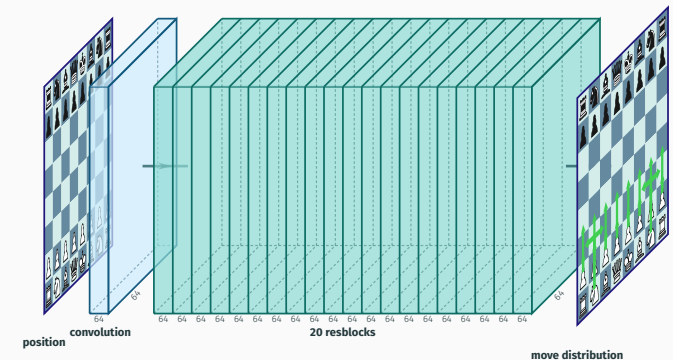
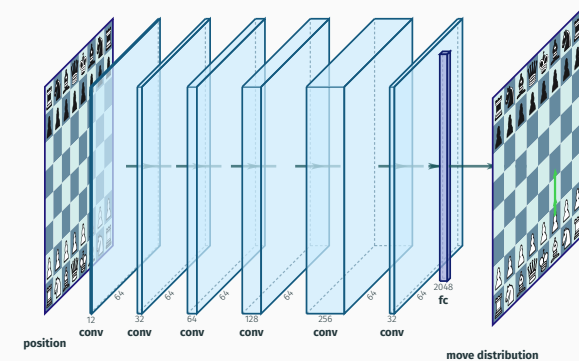
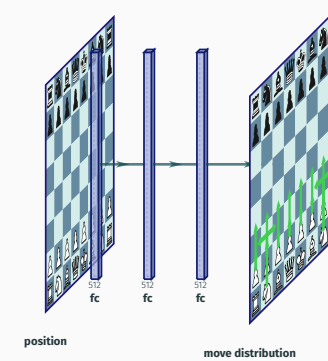
- 3 hidden layers: 512 neurons

CNN

- increasing number of channels:
- $12 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 32$

ResNet

- 20 blocks with 64 channels

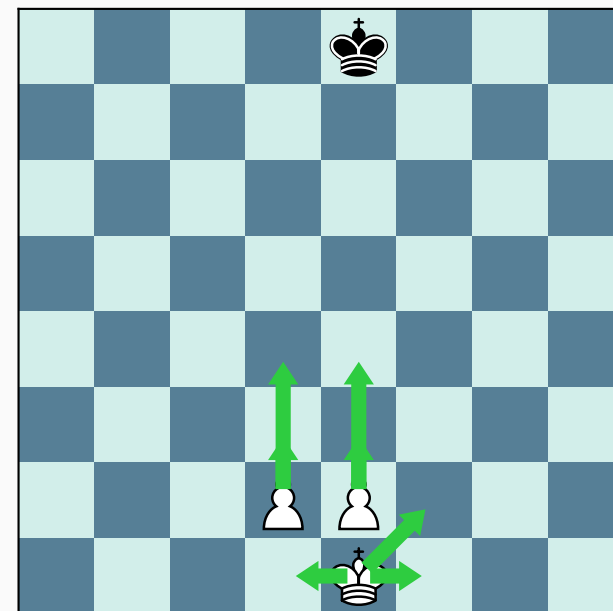


Input

- tensor shape = $(8, 8, 12)$:
 - 8×8 squares
 - 12 pieces: {White, Black} $\times$ {Pawn, Rook, Knight, Bishop, Queen, King}

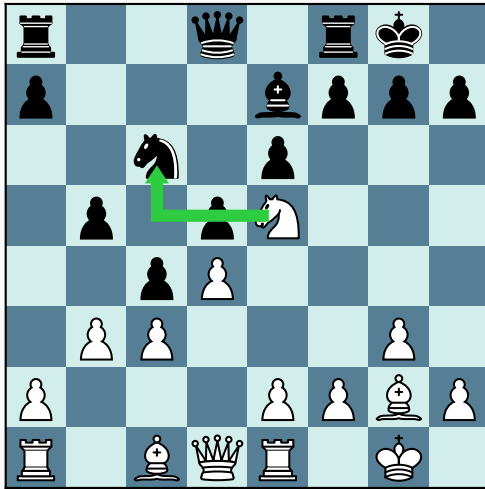
Output

- Move distribution: from square $\rightarrow$ to square
- **But:** Not all moves are legal $\rightarrow$ mask illegal moves



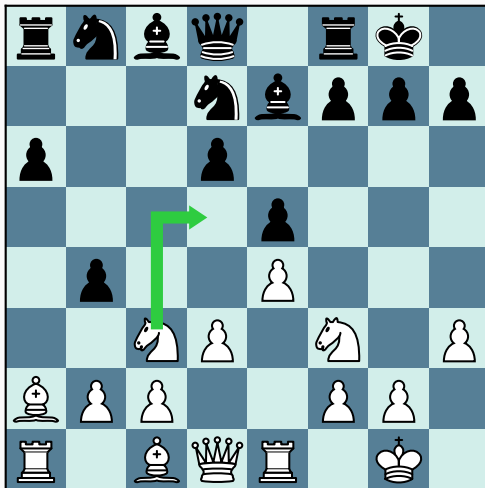
Imitation Learning

Strategy vs Tactics



Tactics

- Short term moves
 - winning a piece
 - checkmate
 - etc.



Strategy

- Long term move:
 - Good piece positioning
 - Pressure on opponent King
 - Control of center
 - etc.

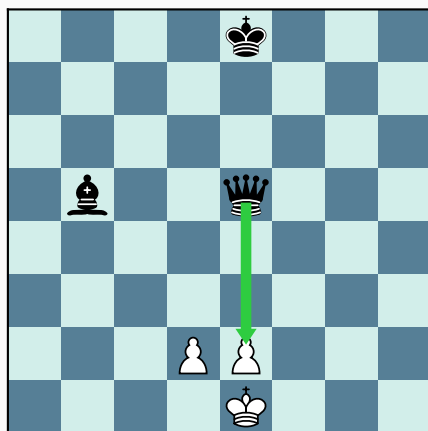
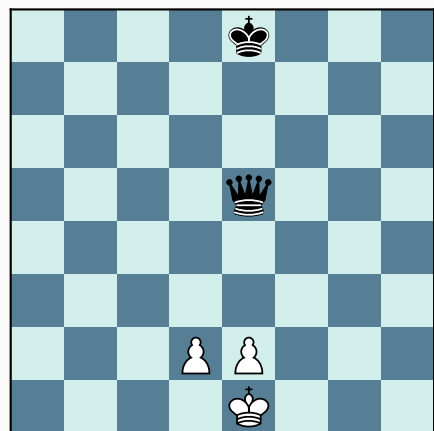
Imitation Learning: Behavior Cloning

Lichess Puzzles [3]

- 6 million positions
- tactical positions

Bias

- no strategic positions
- no positions without tactic

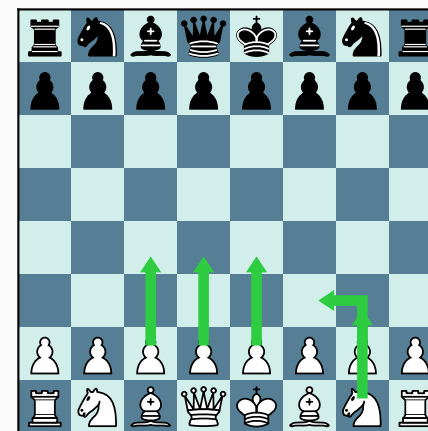
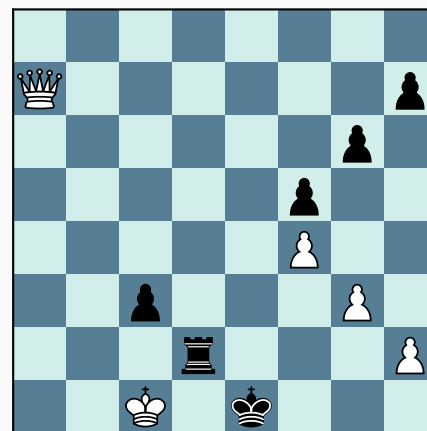


GM games [4]

- chess.com master games
- balance tactics/strategy

Bias & Problems

- high level play
- many resignations
- multiple moves possible



Reinforcement Learning

- **Goal:** Directly optimize a stochastic policy $\pi_{\theta}(a \mid s)$ by maximizing the expected return

$$J(\theta) = E_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T R_t \right].$$

- **Core idea:** Increase the probability of actions that lead to high returns

$$\nabla_{\theta} J(\pi_{\theta}) = E_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T \nabla \log \pi_{\theta}(a_t \mid s_t) G_t \right].$$

- **Algorithm:**
 - Sample trajectories using the current policy π_{θ}
 - Compute the return G_t for each time step
 - Update parameters via gradient ascent

Reward sets

0. **Sparse:** rewards only for true goal
 - reward for win/loss/draw
1. **Medium:** more rewards, but might misalign with goals
 - control center, material win, king safety
2. **Dense:** highest risk of misalignment
 - control outer center, control center, castling, promoting, blunder prevention, material, king safety, give check, win

Training

- Architecture: {FCNN, CNN, ResNet}
 - Pretraining: {Puzzles, Masters} $\times$ {1 Epoch, 10 Epochs}, {Untrained}
 - Reward Set: {NoRL, R_0, R_1, R_2}
 - 1000 batches with 16 games each (REINFORCE)
- 57 models (removing untrained models)

Results

Behavior Cloning: Training Accuracy

dataset	epochs	fc	cnn	resnet
puzzle	1	62.0%	69.9%	74.6%
	10	65.8%	73.3%	81.7%
grand masters	1	31.0%	38.5%	41.5%
	10	31.8%	39.6%	43.7%

- Puzzle better generalizing
- **But** master games: different labels, same position

Behavior Cloning

Evaluation

- All models plays 1000 games against all other model (same architecture)
- Normalized tournament score (win 1 point, draw 0.5 points, loss 0 points)

BC model with RL

dataset	epochs	fc	cnn	resnet
none		0.310	0.251	0.405
puzzle	1	0.615	0.469	0.515
	10	0.562	0.538	0.533
grand masters	1	0.472	0.556	0.510
	10	0.494	0.624	0.513

- Models improve significantly with pretraining

Reward Shaping

RL models with Pretraining

rewards	fc	cnn	resnet
none	0.461	0.419	0.855
r_0	0.520	0.547	0.404
r_1	0.504	0.532	0.405
r_2	0.507	0.486	0.407

RL models without Pretraining

rewards	fc	cnn	resnet
r_0	0.289	0.266	0.408
r_1	0.307	0.235	0.408
r_2	0.335	0.251	0.399

- Depending on the model RL hurts performance
- Sparse rewards generally better
- Example: FC Models with 1 epoch on puzzle get 3800 more points compared to 10 epochs over 72000 games

Conclusion

- We could not find any training strategy across architectures
- Overall denser rewards hurt model performance

Possible Improvements

- **Data Prep:** Unify data in Masters games
- **Architecture:** Hyperparameter tuning on pretrain accuracy
- **BC:** Mix of master games and Puzzles
- **RL:** Longer training
- **RL:** Iterative reward shaping

How does it really play?

Well.... <https://lichess.org/cuRJkeJ1#0>

References

- S. Ibrahim, M. Mostafa, A. Jnadi, H. Salloum, and P. Osinenko,
- [1] “Comprehensive overview of reward engineering and shaping in advancing reinforcement learning applications,” *IEEE Access*, 2024.
 - [2] “Imitation Learning, Stanford University.” [Online]. Available: https://web.stanford.edu/class/cs237b/pdfs/lecture/lecture_10111213.pdf
 - [3] “Lichess chess puzzle dataset.” [Online]. Available: <https://www.kaggle.com/datasets/tianmin/lichess-chess-puzzle-dataset>
 - [4] “All GM Chess Games on Chess.com.” [Online]. Available: <https://www.kaggle.com/datasets/dimitrioskourtikakis/gm-games-chesscom>